

Predictive Maintenance – Chiller System

STANDARD OPERATING PROCEDURE (SOP)

Purpose

This Standard Operating Procedure (SOP) defines the mandatory process for responding to predictive maintenance alerts related to Industrial Chiller Systems (CHLR_1) used for primary temperature cooling. The objective is to ensure safe, consistent intervention to prevent compressor failure, refrigerant loss, or catastrophic loss of cooling capacity.

Scope and Applicability

This SOP applies to all Chiller Systems (CHLR_1, CHLR_2) located in the Central Utility Plant. This procedure must be followed by authorized maintenance personnel whenever predictive interventions are triggered by the AI Engine and delivered through the Smart Action Pack via the M365 Copilot interface.

Roles and Responsibilities

- **HVAC/Refrigeration Technician** - Responsible for executing all physical inspection, diagnostic, and corrective steps in this SOP.
- **Site Supervisor** - Responsible for verifying task completion, validating system stabilization, and approving work order closure.
- **Reliability Engineer** - Responsible for monitoring post-intervention telemetry, adjusting predictive alert thresholds, and optimizing the AI logic.

Overview

This SOP details the mandatory actions required when a predictive maintenance alert indicates a potential restriction in chilled water flow or abnormal compressor vibration within the Chiller System. It outlines the steps to safely isolate the unit, clear flow restrictions, and restore normal cooling operations.

Trigger Conditions (Entry Criteria)

This SOP is initiated when all the following conditions are met:

- Chilled water flow rate drops below baseline thresholds for a sustained period exceeding 15 minutes.

- Compressor vibration levels show unstable oscillation patterns or exceed safe operational limits.
- A Smart Action Pack notification is issued by the AI Engine.

Health & Safety Rules

- Standard site Personal Protective Equipment (PPE) must be worn: safety glasses, steel-toe boots, and impact-resistant gloves.
- When inspecting refrigerant lines, a calibrated portable refrigerant leak detector must be actively used.
- If working near the chilled water pipes or compressor housing, extreme caution must be taken to avoid thermal hazards (cold burns).
- A spill containment kit and appropriate ventilation protocols must be staged nearby before any fluid lines are accessed.

Required Tools

- Mobile device with M365 Copilot and D365 applications installed.
- Calibrated ultrasonic flow meter (for secondary manual verification).
- Portable refrigerant leak detector.
- Vibration analyzer pen.
- Standard insulated hand tools and pipe wrenches.

Step-by-Step Execution Process

- **1. Preparation:** Acknowledge the dispatch from the Smart Action Pack and review the AI-generated predictive timeline and anomaly summary on your mobile device.
- **2. Locating the Asset:** Locate CHLR_1 inside the Central Utility Plant using the asset routing details provided in the ticket.
- **3. Initial Inspection:** Conduct a visual and auditory check of the chiller. Use the portable leak detector along the compressor housing and condenser lines to rule out a hazardous refrigerant leak before proceeding.
- **4. Power Isolation:** Apply strict Lockout/Tagout (LOTO) protocols to the primary local disconnect switch for the chiller unit to safely cut off electrical power.
- **5. Fluid Isolation:** Isolate the chilled water loop by closing the primary upstream and downstream butterfly valves.
- **6. Component Check & Correction:** Open the primary chilled water Y-strainer. Inspect for and remove any accumulated scale, debris, or biological fouling that is restricting water flow. Inspect the compressor mounts for degraded isolation pads causing the vibration alert.

- **7. System Restart:** Reseal the strainer housing, open the isolation valves to restore fluid flow, remove the LOTO equipment, and re-energize the chiller.
- **8. Observation:** Initiate a test cycle. Monitor the localized telemetry dashboard to ensure the chilled water flow rate stabilizes at its designed setpoint and that compressor vibration levels have fallen safely below the original alert threshold.

Exit Criteria

This SOP may be considered complete only when all of the following conditions are met:

- Chilled water flow rate has stabilized within normal operational parameters.
- Compressor vibration levels are verified to be below the alert threshold.
- The chiller has operated continuously for a minimum 15-minute observation period without generating secondary faults.
- No refrigerant leaks are detected.

Exception Handling and Escalation

If flow rates remain restricted or vibration levels do not stabilize after the strainer is cleared:

- Do not return the chiller to active facility service.
- Maintain lockout conditions.
- Escalate immediately to the Site Supervisor and the Reliability Engineering team for advanced diagnostics (e.g., internal compressor inspection or pump impeller check).
- If a refrigerant leak is detected at any point, evacuate the immediate area, trigger the local mechanical room alarm, and follow hazardous gas protocols.

Post-Job Audit and Verification

The following evidence is mandatory and must be uploaded to the D365 work order record:

- Photographic evidence of the cleared strainer.
- Post-intervention manual vibration measurement reading.
- Confirmation of flow rate stabilization.

All checklist items must be digitally signed off within the M365 Copilot interface. Work orders must not be closed until all evidence is verified and approved by the Site Supervisor.

SOP Governance

SOP Owner:	Reliability Engineering Team
Version:	1.0
Approval Authority:	Head of Maintenance
Effective Date:	05/05/2026
Review Cycle:	Annual or upon system change